

Milestone 11 — “4.0.0 release scope”: open issues

Executive overview · spec-kitty/spec-kitty · snapshot 2026-09-28 (main @ b7ada760) · [milestone link](#)

41 open issues (101 already closed)	10 P0 — release-blocking	29 P1	6 epics (multi-mission scope)
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Key takeaways

- **The release gate is red.** 5 of the 10 P0s are CI or test reds ([#5045](#), [#5049](#), [#5169](#), [#5172](#), [#5044](#)). Under the charter's rule that red CI means no release, these block 4.0.0 whatever product scope is chosen.
- **The other P0s are product defects**, and consumers see all of them: Windows skill-file corruption ([#4998](#)), a stale shipped guideline ([#5202](#)), a lane-destruction guard gap ([#5115](#)), decisions erased by repair ([#4919](#)) and lost review verdicts ([#3044](#)).
- **The recurring theme is "silent and exit 0".** Across clusters, commands destroy user data ([#4933](#), [#4940](#)), land wrong results ([#4900](#), [#5046](#), [#2991](#)) or erase state ([#4919](#)) while reporting success. This is the most damaging class for consumer trust.
- **6 epics** ([#1619](#), [#1676](#), [#3044](#), [#4514](#), [#4915](#), [#5001](#)) are architecture-scale. They frame the work but should not be taken on whole before 4.0.0.
- **Next step:** a four-agent research squad is checking each issue against the live code on main. A proposal for one coherent, consumer-focused mission slice will follow.

By cluster

CLUSTER	OPEN	P0	EPICS
Planning-artifact authoring & requirement traceability	11	0	1
User-data preservation, install/upgrade & Windows	7	2	1
Merge / coordination integrity & runtime topology	13	3	3
CI health, test baseline & CLI performance	10	5	1

1. Planning-artifact authoring & requirement traceability

11 issues · 0 P0

What consumers hit while running specify → plan → tasks. Requirement IDs get miscounted or dropped, the error messages don't help, and the workflow assumes a Python codebase. Three of these issues lay the groundwork for structured, CLI-owned planning data; they are bigger than a single fix.

ISSUE	PRI	WHAT IT IS	IMPACT
#5254	P1	The software-dev discovery prompt asks for research artifacts that the research CLI cannot create.	Blocks workflow
#3519	P1	finalize-tasks counts another spec's requirement IDs as this mission's own; map-requirements rejects SC- and letter-suffixed IDs.	Silent wrong result
#2991	P1	finalize-tasks silently drops SC-### success criteria from requirement refs.	Silent wrong result
#2066	P1	When a WP's requirement refs don't match the spec, the diagnostics are opaque and there is no way to see the parsed FR set.	Friction
#2644	P1	Re-finalizing a task graph leaves stale WPCreated/TasksCompleted events behind.	Silent wrong result
#3473	P1	--self-review-fallback reports its force-count inverted.	Misleading output
#2330	P1	The software-dev workflow assumes a Python/pytest layout, which causes friction in Go and other non-Python repos.	Friction

ISSUE	PRI	WHAT IT IS	IMPACT
#1676	P1	Epic: deterministic, structured authoring for planning artifacts.	Groundwork (epic)
#637	P1	Make specify checklist state CLI-owned structured data instead of LLM-written markdown.	Groundwork
#428	P1	Move planning artifacts to JSON-owned state with validation and explainability.	Groundwork
#425	P1	Define canonical JSON schemas for structured planning artifacts.	Groundwork

2. User-data preservation, install/upgrade & Windows

7 issues · 2 P0

Flows that change files in a consumer's repo (skill install, agent config sync, merge) can silently destroy user-authored content and still exit 0. Windows checkouts get corrupted skill files. The shipped doctrine contains a stale duplicate that the runtime serves.

ISSUE	PRI	WHAT IT IS	IMPACT
#4998	P0	On a CRLF checkout, skill install prepends a second frontmatter block to every SKILL.md; doctor reports 275 drifts and upgrade --fix can't repair them.	Corruption (Windows)
#5202	P0	Built-in software-dev guidelines.md exists twice and the copies have drifted; the runtime serves the stale one.	Silent wrong result
#4915	P1	Epic: any flow that changes files must never silently destroy user-authored files.	Data loss (epic)
#4940	P1	agent config sync --sync-hooks replaces a non-UTF-8 .claude/settings.json, losing the user's permissions, env and hooks with no backup. It still exits 0.	Data loss
#4933	P1	merge silently discards uncommitted edits to any tracked file named meta.json, then exits 0.	Data loss
#4930	P1	Windows verification report for 4.0.0rc4 and main (a tracking task).	Verification
#4927	P3	On Windows, upgrade --dry-run still reports 184 phantom repairs.	Misleading output (Windows)

3. Merge / coordination integrity & runtime topology

13 issues · 3 P0

Defects in implement, merge/consolidate, decision moments and review verdicts. Several can land the wrong tree, lose work or erase a decision while exiting 0. Many affect only the coordination topologies (coord / lanes_with_coord). Three architectural epics (5001, 3044, 1619) frame the space.

ISSUE	PRI	WHAT IT IS	IMPACT
#5115	P0	Persist a lane work-tip SHA so the destroyed-lane guard also covers non-coord topologies.	Data-loss guard
#4919	P0	The documented defer → resolve Decision Moment writes two resolved events; doctor says clean, and --repair silently erases the decision.	Silent data loss
#3044	P0	Epic: a negative review verdict must survive to its consumer.	Integrity (epic)
#5001	P1	Epic: a terminus command must never exit 0 while destroying committed work or landing the wrong tree.	Integrity (epic)
#4905	P1	coord / lanes_with_coord: implement commits the WP task file onto the coordination branch, so every later WP's lane allocation fails.	Blocks workflow
#4900	P1	merge prints 'Assigned mission_number=N', but the merge driver keeps null, so every mission ends up numbered 1.	Silent wrong result

ISSUE	PRI	WHAT IT IS	IMPACT
#5046	P1	A canceled WP that committed code to a mixed write-scope lane can still ship its content.	Wrong tree lands
#5255	P1	In coord topology, the runtime issues a WP step with the task body missing.	Blocks workflow
#3449	P1	Missions created with --owned-checkout can't be reached by the runtime loop (status, setup-plan and next resolve to primary).	Blocks workflow
#3097	P1	Post-merge review recovery can't backfill baseline_merge_commit for existing missions.	Recovery gap
#2618	P1	Unify flatten_mission() into one canonical seam (doctor --fix and mission close --discard diverge).	Tech debt
#1619	P1	Epic: unify the mission execution context across the coord, main and lane topologies.	Architecture (epic)
#5047	P2	The terminus mixed-lane test fixture can't build the repros needed to cover #5046 red-first.	Test infra

4. CI health, test baseline & CLI performance

10 issues · 5 P0

Mostly internal, but these set the release gate: red CI means no release. Four nightly suites are red and there are known pre-existing test reds. The two performance issues are the only consumer-visible part of this cluster: slow start-up on every CLI command.

ISSUE	PRI	WHAT IT IS	IMPACT
#5045	P0	Nightly suite red: integration.	Release gate
#5049	P0	Nightly suite red: performance.	Release gate
#5169	P0	Nightly suite red: interpreter-3.13 shard 3.	Release gate
#5172	P0	Nightly suite red: interpreter-3.13 shard 4.	Release gate
#5044	P0	24 pre-existing tests/integration reds on main (the terminus reconciliation-marker fixture drifted).	Release gate
#4916	P1	20 pre-existing Python 3.11 fast-tier test failures.	Test baseline
#3143	P1	The Windows job and local pytest runs get no per-test timeout.	Test infra
#4514	P1	Epic: systemic CLI runtime performance improvements.	Performance (epic)
#4517	P1	Every CLI invocation, even --help and --version, eagerly imports the full events library and command graph.	Consumer latency
#4516	P1	Doctrine and config assets are re-parsed in full on every lookup, with no caching.	Consumer latency

Priorities come from the issues' GitHub labels. Impact tags are a preliminary read of each issue's own description; the research squad has not yet verified them against the code. Issue numbers link to github.com/spec-kitty/spec-kitty.